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ADVANCED PROTOCOL

# CARBOHYDRATE CYCLING

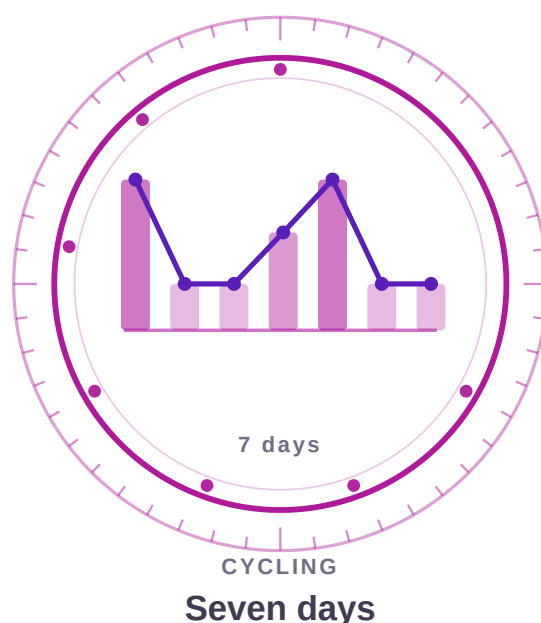
Alternate high, medium and low with your training to improve body composition **without** losing performance.

4 WEEKS

GRAMS PER KG

3 FULL MENUS

REFEEDS



3

DAY TYPES

28

DAYS PLANNED

18

DETAILED MEALS

8

PAGES

## WHAT CARBOHYDRATE CYCLING IS

It is a strategy that **alternates carbohydrate intake** with each day's real demand: plenty of fuel when you train hard, little when you rest. No magic and no accelerated metabolism — simply putting the energy where it is needed.

### WHO IT DOES WORK FOR

- You have at least 6 months of consistent training
- You already know how to count macros and weigh food
- You want to cut without losing strength on the main lifts
- Your week has clearly different days in terms of intensity

### WHO IT IS NOT FOR YET

- You are starting out: master a plain surplus or deficit first
- You do not log what you eat accurately
- You have a history of eating disorders
- Your schedule changes every week and you cannot plan

## THE THREE DAY TYPES

| DAY TYPE   | CARBOHYDRATES | APPROX. CALORIES | WHEN TO USE IT                |
|------------|---------------|------------------|-------------------------------|
| HIGH · H   | 300–380 g     | 2.800–3.100      | Heavy training: legs or back  |
| MEDIUM · M | 200–250 g     | 2.400–2.600      | Moderate training: upper body |
| LOW · L    | 80–120 g      | 1.900–2.200      | Rest or light cardio (LISS)   |

The table values are for a reference athlete of 75 kg. On page 3 you convert them to **your** body weight.

## WHY IT WORKS

- **Insulin sensitivity:** after a hard session, muscle takes up glucose far more efficiently. The high day's carbohydrates go to glycogen, not to fat tissue.
- **Performance protected:** the demanding days arrive with full stores, so you keep your loads while the weekly average stays in deficit.
- **Adherence:** psychologically it is far easier to live with than a flat, monotonous deficit for eight weeks.

### WHAT CYCLING DOES NOT DO

It does not burn fat by itself. Fat loss still depends on the **weekly calorie balance**. Cycling merely distributes those calories better so that you perform more and keep muscle while the balance does its work.

# CONVERT THE PROTOCOL TO YOUR WEIGHT

Generic figures only serve someone who weighs exactly the same as the reference athlete. These are the protocol's real ratios: multiply them by your kilos and you have your own version.

| DAY TYPE | CARBOHYDRATES | PROTEIN  | FATS     | LOGIC                                  |
|----------|---------------|----------|----------|----------------------------------------|
| HIGH     | 4.5–5.5 g/kg  | 2.2 g/kg | 0.8 g/kg | High carbs, low fat                    |
| MEDIUM   | 3.0–3.5 g/kg  | 2.4 g/kg | 1.0 g/kg | The balance point                      |
| LOW      | 1.0–1.5 g/kg  | 2.6 g/kg | 1.2 g/kg | Fat rises, protein protects the muscle |

THE PROTOCOL'S KEY RULE

Protein **rises** when carbohydrates fall, and fats do the opposite of carbs. That way calories do not collapse on low days and you protect muscle mass exactly when the risk of losing it is highest.

## WORKED EXAMPLE · 75 KG ATHLETE

| DAY                              | CARBS | PROTEIN | FATS | CALORIES   |
|----------------------------------|-------|---------|------|------------|
| High (5.0 g/kg)                  | 375 g | 165 g   | 60 g | 2.700      |
| Medium (3.2 g/kg)                | 240 g | 180 g   | 75 g | 2.355      |
| Low (1.3 g/kg)                   | 98 g  | 195 g   | 90 g | 1.982      |
| Weekly average (3 H · 2 M · 2 L) | 243 g | 177 g   | 73 g | 2,386 kcal |

## YOUR TABLE — WORK IT OUT ONCE AND USE IT FOR 4 WEEKS

| DAY    | CARBS (G/KG × _____ KG) | PROTEIN | FATS  | CALORIES |
|--------|-------------------------|---------|-------|----------|
| High   | _____                   | _____   | _____ | _____    |
| Medium | _____                   | _____   | _____ | _____    |
| Low    | _____                   | _____   | _____ | _____    |

Remember: 1 g of carbohydrate = 4 kcal · 1 g of protein = 4 kcal · 1 g of fat = 9 kcal. Fibre is already included in the carbohydrate count.

## 4-WEEK CALENDAR

| WEEK   | MON | TUE | WED | THU | FRI | SAT | SUN |
|--------|-----|-----|-----|-----|-----|-----|-----|
| Week 1 | A   | M   | A   | B   | A   | M   | B   |
| Week 2 | A   | B   | A   | M   | A   | B   | B   |
| Week 3 | M   | A   | B   | A   | M   | A   | B   |
| Week 4 | A   | M   | A   | B   | A   | A   | M   |

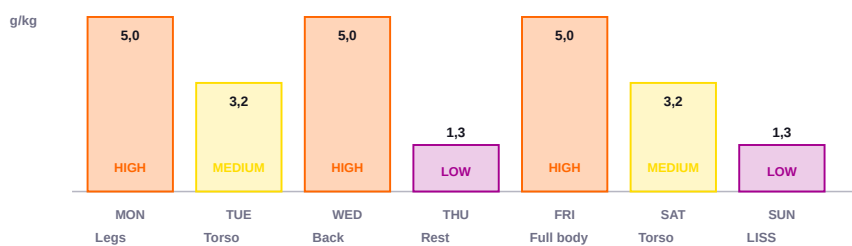
### ADAPT THE PATTERN TO YOUR WEEK, NOT THE OTHER WAY ROUND

This calendar assumes heavy training on Monday, Wednesday and Friday. If your leg day is Tuesday, move the HIGH day to Tuesday. The rule that never changes is: **high day = day of greatest training volume**.

## HOW TO PAIR EACH DAY WITH YOUR TRAINING

| YOUR SESSION             | DAY TYPE | HOW THE CARBOHYDRATES ARE SPLIT                          |
|--------------------------|----------|----------------------------------------------------------|
| Heavy legs or back       | HIGH     | 50 % before training · 30 % after · 20 % rest of the day |
| Chest, shoulders or arms | MEDIUM   | 40 % before · 30 % after · 30 % rest                     |
| Full body or hybrid      | HIGH     | 45 % before · 35 % after · 20 % rest                     |
| LISS cardio or mobility  | LOW      | Split as you like, prioritise vegetables and protein     |
| Full rest                | LOW      | Split as you like, a fattier dinner for satiety          |

FIG. 1 THE CYCLING WEEK AT A GLANCE



The rule that does not change: the high day goes where the training volume is highest. The weekly average still lands in deficit even

Each bar's height = grams of carbohydrate per kilo of body weight. Compare this figure with the table on the previous page before setting your own week.

# HIGH-CARBOHYDRATE DAY

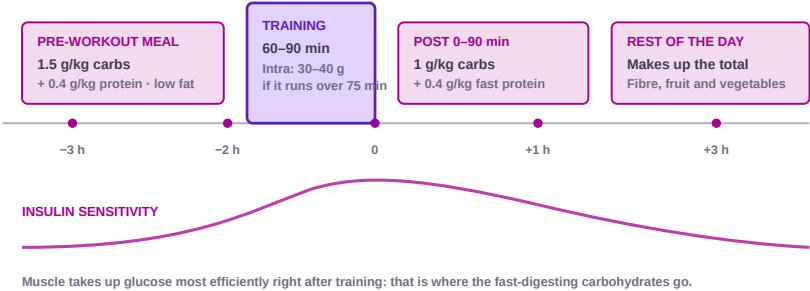
|               |         |      |       |
|---------------|---------|------|-------|
| 366 g         | 205 g   | 62 g | 2.850 |
| CARBOHYDRATES | PROTEIN | FATS | KCAL  |

| MEAL         | FOODS                                                  | CARBS | PROT. | KCAL  |
|--------------|--------------------------------------------------------|-------|-------|-------|
| Breakfast    | Oats 100 g + banana + 3 egg whites                     | 85 g  | 28 g  | 520   |
| Pre-workout  | Cooked rice 200 g + chicken 200 g + tomato             | 92 g  | 48 g  | 680   |
| Post-workout | Protein shake + rice pudding 100 g                     | 55 g  | 42 g  | 450   |
| Lunch        | Pasta 150 g + salmon 180 g + salad                     | 88 g  | 45 g  | 720   |
| Dinner       | Sweet potato 200 g + chicken breast 180 g + vegetables | 46 g  | 42 g  | 480   |
| Day total    | —                                                      | 366 g | 205 g | 2.850 |

## HIGH-DAY TIMING

| WINDOW          | TARGET             | PRACTICAL DETAIL                                                           |
|-----------------|--------------------|----------------------------------------------------------------------------|
| 2 h before      | Fill the stores    | The day's biggest carbohydrate meal, low in fat so it digests fast         |
| Intra-workout   | Hold the intensity | Only if the session runs over 75 min: 30–40 g of liquid carbohydrate       |
| 0–90 min after  | Replenish glycogen | Fast-digesting carbohydrate + 40 g of protein                              |
| Rest of the day | Make up the total  | Wholegrain sources, fruit and vegetables to reach fibre and micronutrients |

FIG. 2 THE HIGH DAY'S PERI-WORKOUT WINDOW



The grams are in g/kg of body weight, so they scale by themselves. The lower curve is the reason for the whole protocol.

### DROP THE FAT WHEN THE CARBS GO UP

The high day is not a day for eating more of everything. Fats drop to 0.8 g/kg precisely to make caloric room for the carbohydrates. If you keep fat high and raise carbs as well, you have created a surplus by accident.

## MEDIUM-CARBOHYDRATE DAY

**240 g**

CARBOHYDRATES

**202 g**

PROTEIN

**76 g**

FATS

**2.500**

KCAL

| MEAL         | FOODS                                                               | CARBS | PROT. | KCAL  |
|--------------|---------------------------------------------------------------------|-------|-------|-------|
| Breakfast    | Oats 60 g + 2 eggs + 2 egg whites + blueberries                     | 40 g  | 26 g  | 420   |
| Pre-workout  | Rice 120 g dry + chicken 180 g + sautéed vegetables                 | 92 g  | 44 g  | 660   |
| Post-workout | Protein shake + large banana                                        | 30 g  | 27 g  | 250   |
| Lunch        | Sweet potato 200 g + salmon 150 g + green salad                     | 40 g  | 35 g  | 490   |
| Dinner       | Chicken breast 180 g + quinoa 40 g + broccoli + 1 tbsp of olive oil | 30 g  | 44 g  | 430   |
| Before bed   | Cottage cheese 200 g + walnuts 15 g                                 | 8 g   | 26 g  | 250   |
| Day total    | —                                                                   | 240 g | 202 g | 2.500 |

Menu calculated for a 75 kg athlete. Scale the amounts in proportion to your weight using the table on page 3.

### THE MEDIUM DAY IS THE MOST USEFUL OF THE THREE

- It is your **default day**: when in doubt about which type of day it is, choose medium.
- It fits 70 % of real sessions: upper body, technique or moderate volume.
- It lets you train well without piling up the fatigue of repeated low days.
- It is the day you will use to **maintain** once the protocol is over.

#### VALID SWAPS FOR THE MEDIUM DAY

Sweet potato ↔ potato ↔ brown rice · Salmon ↔ mackerel ↔ fresh tuna · Cottage cheese ↔ 0 % Greek yoghurt ↔ casein · Quinoa ↔ wholewheat couscous ↔ bulgur. Keep the grams of carbohydrate, not the weight of the food.

## LOW-CARBOHYDRATE DAY

**75 g**

CARBOHYDRATES

**219 g**

PROTEIN

**92 g**

FATS

**2.160**

KCAL

| MEAL       | FOODS                                       | CARBS | PROT. | KCAL  |
|------------|---------------------------------------------|-------|-------|-------|
| Breakfast  | Omelette of 4 egg whites + 2 eggs + avocado | 6 g   | 36 g  | 380   |
| Snack 1    | Greek yoghurt + nuts                        | 12 g  | 22 g  | 280   |
| Lunch      | Salmon 220 g + large salad + olive oil      | 18 g  | 48 g  | 520   |
| Snack 2    | Whey protein + almonds 30 g                 | 10 g  | 35 g  | 340   |
| Dinner     | Chicken breast 200 g + broccoli + spinach   | 15 g  | 50 g  | 380   |
| Before bed | Cottage cheese 200 g + chia seeds           | 14 g  | 28 g  | 260   |
| Day total  | —                                           | 75 g  | 219 g | 2.160 |

### HOW TO GET THROUGH A LOW DAY WELL

| PROBLEM                 | CAUSE                                           | SOLUTION                                                          |
|-------------------------|-------------------------------------------------|-------------------------------------------------------------------|
| Mid-afternoon hunger    | Too little food volume                          | Double the vegetables: they add satiety at almost no calorie cost |
| Headache                | Lack of sodium and water                        | Add salt to your meals and add 500 ml of water                    |
| Strong sugar craving    | Glucose dip and habit                           | Sugar-free jelly, herbal tea or 20 g of 90 % chocolate            |
| Feeling flat in the gym | You have put the low day next to a hard session | Reorder the week: the low day belongs with rest or easy cardio    |
| Constipation            | Less fibre when cereals go down                 | Leafy greens, chia, psyllium and 2.5 L of water                   |

#### NEVER CHAIN MORE THAN TWO LOW DAYS IN A ROW

Three or more consecutive low days sink performance, sleep quality and mood, and send the odds of a binge through the roof. If the calendar forces it, turn the third into a medium day.

## WHERE EVERY NUMBER COMES FROM

Every number in this document comes from sources that are public, free and checkable. Not one figure is invented, rounded for convenience or copied from another popular publication. This page exists so that you can verify it yourself.

### WHY THESE SOURCES AND NO OTHERS

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| WHAT YOU ARE LOOKING FOR     | WHAT YOU WILL FIND                                                           | WHERE                                                     |
|------------------------------|------------------------------------------------------------------------------|-----------------------------------------------------------|
| Reference intakes            | RDA, AI and tolerable upper intake level for every vitamin and mineral       | <a href="https://ods.od.nih.gov">ods.od.nih.gov</a>       |
| Food composition             | The full profile of any food, nutrient by nutrient                           | <a href="https://fdc.nal.usda.gov">fdc.nal.usda.gov</a>   |
| Dietary guidelines           | Dietary patterns and limits for added sugars, saturated fat and sodium       | <a href="https://health.gov">health.gov</a>               |
| Physical activity guidelines | The weekly dose, its response curve and the strength requirement             | <a href="https://health.gov">health.gov</a>               |
| Physical readiness doctrine  | FM 7-22: training, nutrition, sleep and mental readiness                     | <a href="https://armypubs.army.mil">armypubs.army.mil</a> |
| Lifting equation             | How much weight is reasonable to lift in each specific situation             | <a href="https://cdc.gov/niosh">cdc.gov/niosh</a>         |
| Food safety                  | Safe internal temperatures and the danger zone                               | <a href="https://foodsafety.gov">foodsafety.gov</a>       |
| Health claims                | What may legally be claimed about a nutrient in Europe, and what was refused | European Commission register                              |

### HOW THIS DOCUMENT CHECKS ITSELF

The reference figures are not typed by hand: they live in a data module that the typesetting program verifies before writing a single line. It checks that the macronutrient ranges can add up to one hundred per cent of energy, that none of the 19 micronutrients has a recommended value above its own ceiling without a note explaining why, and that no safe cooking temperature falls inside the danger zone. If a single check fails, the document is not typeset.

### REFERENCE VALUES ARE POPULATION VALUES

They describe what covers the needs of most healthy adults. They are not an individual prescription, they do not account for disease, pregnancy, breastfeeding or medication, and no publication replaces assessment by a registered health professional. If you take medication, some of the nutrients named here interact with it in clinically relevant ways: do not change an intake or start a supplement without telling whoever prescribes for you.

## REFEEDS, ADJUSTMENTS AND SIGNALS

### WHEN AND HOW TO RUN A REFEED

| SIGNAL                                       | WHAT IT INDICATES                   | ACTION                                       |
|----------------------------------------------|-------------------------------------|----------------------------------------------|
| Weight flat for 2 weeks and strength falling | Empty stores, not a missing deficit | One extra high day at 6 g/kg of carbohydrate |
| Weight dropping more than 1 % a week         | Deficit too aggressive              | Turn a low day into a medium one             |
| Weight up two weeks in a row                 | No real deficit                     | Turn a high day into a medium one            |
| Poor sleep and high resting pulse            | Accumulated fatigue                 | A full week at medium level before resuming  |

### 4-WEEK LOG

| WEEK   | AVERAGE WEIGHT | WAIST | STRENGTH (FEEL 1-10) | ADJUSTMENT APPLIED |
|--------|----------------|-------|----------------------|--------------------|
| Week 1 |                |       |                      |                    |
| Week 2 |                |       |                      |                    |
| Week 3 |                |       |                      |                    |
| Week 4 |                |       |                      |                    |

### CYCLING IS TUNED WITH DATA, NOT FAITH

Give the protocol **three full weeks** before you touch anything, then change it with the scale and the log in front of you. A cycling plan corrected every few days is not a cycling plan: it is noise.

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